

1. FOUNDATIONS BEYOND BASIC PROBABILITY

1.1 Avatar and Notation



Ambient objects. Work on standard Borel spaces $(E, \mathcal{E})$, Polish state spaces, separable Banach spaces $\mathbb{B}$, and filtered probability bases $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$.

Alphabet $\mathcal{F}_t, \mathbb{R}^d, \mathcal{Q}, \mathfrak{g}, \theta$

Remember. The file `tigs/avatar.jpg` is intentionally square. Replace it with your own square avatar without changing the `\LaTeX` command.

1.2 Disintegration and Kernels

Markov kernel. A map $K : E \times \mathcal{F} \rightarrow [0, 1]$ such that $x \mapsto K(x, A)$ is measurable and $A \mapsto K(x, A)$ is a probability measure.

Disintegration $\mu(dx, dy) = \mu_X(dx)K(x, y)$

Doob–Dynkin Y is $\sigma(X)$ -meas. $\Rightarrow Y = g(X)$ a.s.

■ 1.2.1 Regular conditional law $K(x, \cdot) = \mathbb{P}(Y \in \cdot \mid X = x)$ is not a pointwise conditional probability in arbitrary measurable spaces; standard Borel regularity matters.

Warning. Conditioning on null events is not defined by Bayes’ formula; use kernels, versions, or limiting constructions.

1.3 Modes of Weakness

Mode	Memory hook
Weak convergence	Test against $C_b(E)$, then invoke Prokhorov tightness.
Stable convergence	Joint convergence with every bounded variable from the base sigma-field.
Contiguity	$F_n \triangleleft Q_n$ transfers $o_{Q_n}(1)$ events into $o_{P_n}(1)$ events.
Uniform integrability	Converts convergence in probability into L^1 convergence when tails are tight.

Example: Portmanteau To prove $\mu_n \Rightarrow \mu$, control closed sets, open sets, or bounded continuous test functions; choose the form that matches the problem.

2. ASYMPTOTIC STATISTICS AND EXPERIMENTS

2.1 LAN, LAMN, Le Cam

LAN. A statistical experiment is locally asymptotically normal if likelihood ratios around θ_0 admit a quadratic expansion with central sequence Δ_n .

LAN expansion $\ell_{\theta_0+h/\sqrt{n}} - \ell_{\theta_0} = h^\top \Delta_n - \frac{1}{2} h^\top I h + o_p(1)$

Third lemma $\Delta_n \Rightarrow N(0, I)$ under $P_{\theta_0} \Rightarrow \Delta_n \Rightarrow N(Ih, I)$ under $P_{\theta_0+h/\sqrt{n}}$

Remember. Contiguity is the hidden bridge: it permits changing local measures without losing $o_p(1)$ remainders.

2.2 Semiparametric Geometry

Tangent space. The closed linear span of scores along parametric submodels through the nonparametric model.

Efficient influence function. The canonical gradient: the orthogonal projection of any influence function onto the orthocomplement of nuisance tangent directions.

EIF identity $\sqrt{n}(\hat{\psi} - \psi_0) = n^{-1/2} \sum_i \phi_0(Z_i) + o_p(1)$

One-step estimator
Start with a rough estimator $\hat{\eta}$, estimate the influence function $\hat{\phi}$, then correct the plug-in estimator by $n^{-1} \sum_i \hat{\phi}(Z_i)$. Orthogonality makes nuisance error second order.

2.3 Empirical Processes

- Donsker class: $\sqrt{n}(P_n - P) \rightsquigarrow \mathbb{G}_P$ in $\ell^\infty(\mathcal{F})$.
- Bracketing entropy: controls stochastic equicontinuity for nonparametric M-estimators.
- Argmax theorem: turns process convergence into estimator convergence.

Entropy $\int_0^1 \sqrt{\log N_{[]}(\varepsilon, \mathcal{F}, L_2)}$

3. STOCHASTIC ANALYSIS

3.1 Martingales and Compensators

Doob–Meyer. $X_t = M_t + A_t$ for a submartingale under integrability and regularity; A is predictable increasing.

Quadratic variation $[M]_t = M_t^2 - 2 \int_0^t M_s - dM_s$ compensates M^2

Optional stopping $\mathbb{E}[M_\tau] = \mathbb{E}[M_0]$ under bounded/Ül stopping conditions

Warning. A local martingale is not automatically a true martingale; check integrability, Novikov, Kazamaki, or class D .

3.2 Girsanov and Ito

Stochastic exponential. $\mathcal{E}(\theta \cdot W)_t = \exp\{\int_0^t \theta_s dW_s - \frac{1}{2} \int_0^t \|\theta_s\|^2 ds\}$.

Girsanov $d\tilde{\mathbb{P}} = \mathcal{E}(\theta \cdot W)_T d\mathbb{P}$, $\tilde{W}_t = W_t - \int_0^t \theta_s ds$

Ito $df(t, X_t) = (\partial_t f + \mathcal{L}f) dt + \nabla f^\top \sigma dW_t$

■ 3.2.1 Generator For $dX_t = b(X_t) dt + \sigma(X_t) dW_t$, $\mathcal{L}f = b \cdot \nabla f + \frac{1}{2} \text{tr}(\sigma \sigma^\top \nabla^2 f)$.

3.3 Malliavin Snapshot

Malliavin derivative. $D_\theta F$ differentiates a functional $F(W)$ along Cameron–Martin directions.

Clark–Ocone $F = \mathbb{E}[F] + \int_0^T \mathbb{E}[D_t F \mid \mathcal{F}_t] dW_t$

Example: Density criterion Non-degeneracy of the Malliavin covariance often implies smooth densities for diffusion laws.

4. VARIATIONAL AND CONVEX ANALYSIS

4.1 Subdifferentials

Fenchel subgradient. $p \in \partial f(x)$ iff $f(y) \geq f(x) + \langle p, y - x \rangle$ for every y .

Fenchel–Moreau $f = f^{**}$ for proper l.s.c. convex f

Fenchel–Rockafellar $\inf_x f(x) + g(Ax) = \sup_y -f^*(-A^\top y) - g^*(y)$

Remember. Most duals are just conjugates plus an exchange of infimum and supremum; regularity decides whether the exchange is legal.

4.2 Γ-Convergence

Γ-limit. $F_n \xrightarrow{\Gamma} F$ if the liminf inequality and recovery sequence property both hold.

1. liminf: $x_n \rightarrow x \Rightarrow F(x) \leq \liminf_n F_n(x_n)$;
2. recovery: $\exists x_n \rightarrow x$ with $F(x) \geq \limsup_n F_n(x_n)$;
3. equicoercivity then transfers minimizers.

Warning. Γ-convergence is designed for minimization, not point-wise approximation.

4.3 Proximal Geometry

Resolvent $J_{\lambda A} = (I + \lambda A)^{-1}$, A maximal monotone

Moreau envelope $e_\lambda f(x) = \inf_y \{f(y) + \frac{1}{2\lambda} \|y - x\|^2\}$

Object	Why it matters
Proximal map	implicit Euler step for gradient flows in nonsmooth landscapes.
Bregman divergence	replaces Euclidean geometry in mirror descent.
Normal cone	encodes constraints as a set-valued subgradient.

5. OPTIMAL TRANSPORT & INFORMATION GEOMETRY

5.1 Kantorovich Duality

Transport plan. $\pi \in \Pi(\mu, \nu)$ couples marginals μ and ν ; Monge asks for deterministic maps, Kantorovich relaxes to couplings.

Primal $\inf_{\pi \in \Pi(\mu, \nu)} \int c(x, y) d\pi(x, y)$

Dual $\sup_{\varphi + \psi \leq c} \int \varphi d\mu + \int \psi d\nu$

■ 5.1.1 Wasserstein gradient flow Many PDEs can be written as steepest descent of an energy functional in the metric space $(\mathcal{P}_2, W_2)$. Example: JKO scheme $\rho_{k+1} = \arg \min_{\rho} \{ \mathcal{E}(\rho) + \frac{1}{2\tau} W_2^2(\rho, \rho_k) \}$.

5.2 Schrodinger Bridge

Entropic OT. Replace hard transport by a KL-regularized coupling problem; Sinkhorn iterations are alternating Bregman projections.

Entropic primal $\inf_{\pi \in \Pi(\mu, \nu)} \int c d\pi + \varepsilon \text{KL}(\pi \| \mu \otimes \nu)$

Remember. As $\varepsilon \downarrow 0$, entropic OT approaches Kantorovich OT; for positive ε , it is smoother and numerically stable.

5.3 Information Geometry

Natural gradient. Gradient flow under the Fisher–Rao metric rather than the Euclidean metric.

Fisher metric $\langle u, v \rangle_\theta = u^\top I(\theta) v$, $\dot{\theta} = -I(\theta)^{-1} \nabla L(\theta)$

Warning. Parameterization matters for ordinary gradients; natural gradients are invariant under smooth reparameterization.

6. SPECTRAL, OPERATOR, AND LEARNING THEORY

6.1 Spectral Regularization

Compact inverse problem. When singular values $\sigma_j \downarrow 0$, naive inversion amplifies high-frequency noise.

Tikhonov filter $x_\alpha = \sum_j \frac{\sigma_j}{\sigma_j^2 + \alpha} \langle y, u_j \rangle v_j$

Spectral GCN $g_\theta(L)x = U g_\theta(\Lambda) U^\top x$

Chebyshev recurrence
 $T_0(\tilde{L})x = x, T_1(\tilde{L})x = \tilde{L}x$, and $T_k(\tilde{L})x = 2\tilde{L}T_{k-1}(\tilde{L})x - T_{k-2}(\tilde{L})x$.

6.2 PAC-Bayes and Stability

PAC-Bayes skeleton $\mathbb{E}_{h \sim Q} R(h) \lesssim \mathbb{E}_{h \sim Q} \hat{R}(h) + \sqrt{\frac{\text{KL}(Q \| P) + \log(1/\delta)}{n}}$

Algorithmic stability. A learning algorithm generalizes if replacing one sample barely changes its loss.

- **Compression:** prove generalization by describing the learned predictor with few bits.
- **Margin:** convert geometry of separators into risk control.
- **Flatness:** useful only when made parameterization-aware.

6.3 Nonasymptotic Random Matrices

Matrix Bernstein $\|\sum_i X_i\| \lesssim \sqrt{v \log d} + L \log d$ with high probability

Resolvent $G(z) = (H - zI)^{-1}$, $m_n(z) = n^{-1} \text{tr } G(z)$

Example: Local law A typical proof controls resolvent entries down to nearly optimal spectral scales, then transfers to eigenvalue rigidity.

7. NOTATION AND QUICK REFERENCE

7.1 Blocks, Lists, Tables

This line demonstrates `term`, `key`, `code`, and `highlight`. Use visual emphasis sparingly; density is the point of a cheat sheet.

Proof skeleton for impossible-looking problems
Reduce the claim to compactness, monotonicity, orthogonality, or duality. Then choose one of: Prokhorov + Skorokhod, Hahn–Banach separation, martingale localization, spectral truncation, or epi/Γ-convergence.

Need	Command	Hint
Big title	<code>\cheatsection</code>	numbered bar
Small title	<code>\topic</code>	compact block
Auto table	<code>\cheatablex</code>	use <code>\column</code>
Image	<code>\colimage</code>	square avatar works

■ 7.1.1 中文兼容示例 若要证明估计量的渐近正态性,通常先建立局部二次展开,再用随机等度连续性控制余项,最后通过 argmax 连续映射定理得到极限分布。

Final mnemonic tightness + identifiability + stability $\Rightarrow$ limit theorem